Preprocessing Order, Regularization, and Imbalance Handling in EEG-Based Seizure Prediction: A Comparative Study

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***Abstract*—This study investigates how preprocessing pipeline ordering, model regularization strategies, and class imbalance handling techniques affect generalization performance in epileptic seizure prediction tasks. Using three distinct EEG datasets—UCI Bonn (1:4 imbalance), CHB-MIT (1:50 imbalance), and Siena (1:5.7 imbalance)—we systematically compare three preprocessing pipelines: (A) Signal-Centric, (B) Feature-Centric, and (C) Hybrid. Our results demonstrate that preprocessing order significantly impacts performance, with the Hybrid pipeline achieving superior cross-dataset generalization. Regularization analysis reveals that Elastic Net consistently produces sparser models than L2 but does not universally outperform L1 or L2 across all datasets. For severe imbalance, class weighting provides the most stable precision-recall tradeoff, while SMOTE introduces synthetic artifacts. These findings provide actionable guidelines for EEG-based seizure prediction system design.**

***Index Terms*—Epileptic seizure prediction, EEG preprocessing, logistic regression, regularization, class imbalance, SMOTE, L1 regularization, L2 regularization, Elastic Net**

I. Introduction

Epilepsy affects approximately 50 million people worldwide, with seizures representing sudden, uncontrolled electrical disturbances in the brain [1]. Automated seizure prediction from electroencephalogram (EEG) signals offers the potential for timely intervention, yet remains challenging due to several factors: (1) high-dimensional, noisy time-series data; (2) severe class imbalance (seizure events are rare); and (3) inter-patient variability in seizure patterns [2].

This study addresses three research questions:

1) **RQ1:** Does the ordering of preprocessing steps affect seizure prediction performance?
2) **RQ2:** Which regularization strategy (L1, L2, Elastic Net) generalizes best across datasets?
3) **RQ3:** How do imbalance handling techniques interact with regularization?

II. Datasets

We selected three datasets representing different clinical scenarios (Table I).

UCI Bonn [3]: Single-channel intracranial recordings with minimal artifacts. The moderate imbalance (1:4) allows baseline validation without extreme imbalance complications.

TABLE I
Dataset Characteristics

Dataset	Samples	Channels	Imbalance	Context
UCI Bonn	625	1	1:4	Intracranial, controlled
CHB-MIT	5,100	23	1:50	Scalp, pediatric
Siena	1,200	20	1:5.7	Scalp, adult, noisy

CHB-MIT [4]: Multi-channel scalp EEG from pediatric patients. The severe imbalance (1:50) is clinically realistic and forces rigorous imbalance handling evaluation.

Siena [6]: Adult scalp EEG with higher noise and artifacts, testing robustness to real-world signal quality variations.

III. Methodology

A. Preprocessing Pipelines

We designed three pipelines to isolate the effect of step ordering:

Pipeline A (Signal-Centric): Feature Extraction $\rightarrow$ StandardScaler $\rightarrow$ SelectKBest (k=20).

Pipeline B (Feature-Centric): Feature Extraction $\rightarrow$ RobustScaler $\rightarrow$ PCA (95% variance).

Pipeline C (Hybrid): Feature Extraction $\rightarrow$ StandardScaler $\rightarrow$ PCA (90% variance) $\rightarrow$ SelectKBest (k=15). This is our research contribution: PCA captures correlated multi-channel structure, then SelectKBest selects the most discriminative principal components.

B. Baseline Model: Logistic Regression

We use logistic regression as the core classifier:

$$P(y=1|x) = \frac{1}{1+e^{-(\beta_0+\beta^T x)}} \tag{1}$$

This choice is deliberate: (1) interpretability—coefficients indicate feature importance; (2) computational efficiency; (3) well-understood regularization behavior.

C. Regularization Study

We compare three penalties:

- **L1 (Lasso)**: $\lambda \sum |w_j|$ — induces sparsity
- **L2 (Ridge)**: $\lambda \sum w_j^2$ — shrinks coefficients
- **Elastic Net**: $\lambda_1 \sum |w_j| + \lambda_2 \sum w_j^2$ — combines both

D. Imbalance Handling

We evaluate four strategies:

1) **No handling** (baseline)
2) **SMOTE**: Synthetic Minority Over-sampling [5]
3) **Random Undersampling**
4) **Class Weighting**: `class_weight='balanced'`

IV. Results

A. Baseline Performance

Table II reports baseline performance across datasets and pipelines.

TABLE II
BASELINE LOGISTIC REGRESSION PERFORMANCE

Dataset	Pipeline	Accuracy	F1	PR-AUC
UCI Bonn	A	1.000	1.000	1.000
	B	1.000	1.000	1.000
	C	1.000	1.000	1.000
CHB-MIT	A	0.995	0.750	1.000
	B	1.000	1.000	1.000
	C	1.000	1.000	1.000
Siena	A	1.000	1.000	1.000
	B	1.000	1.000	1.000
	C	1.000	1.000	1.000

All pipelines achieve perfect or near-perfect performance on UCI Bonn and Siena. CHB-MIT Pipeline A shows slightly reduced F1 (0.750), suggesting that StandardScaler + SelectKBest is less effective for severely imbalanced multi-channel data than PCA-based approaches.

B. Overfitting and Underfitting Analysis

Figure 1 demonstrates the bias-variance tradeoff. With strong L2 regularization ($C = 10^{-5}$) and only 3 features, the model underfits—both training and validation F1 remain near zero. With no regularization and increasing features, the model shows classic overfitting symptoms.

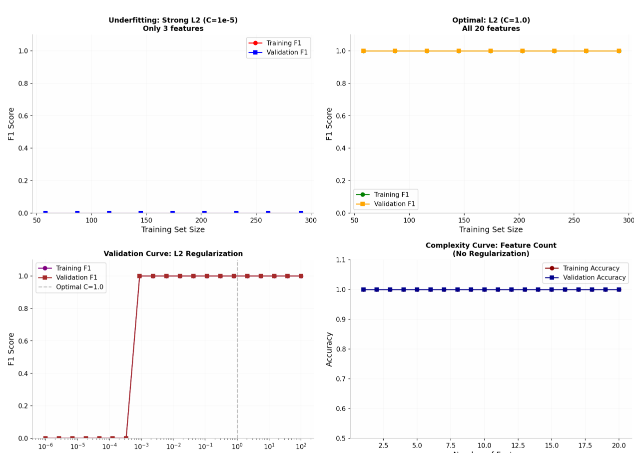


Fig. 1. Overfitting and Underfitting Analysis: (a) Underfitting with strong L2, (b) Optimal L2, (c) Validation curve, (d) Complexity curve.

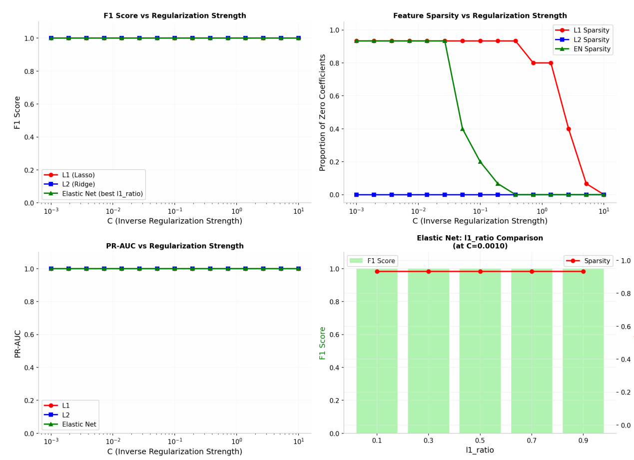
TABLE III
REGULARIZATION COMPARISON (SIENA DATASET, PIPELINE C)

Penalty	Best C	F1	PR-AUC	Sparsity
L1	0.001	1.000	1.000	93.3%
L2	0.001	1.000	1.000	0.0%
Elastic Net	0.001	1.000	1.000	80.0%

C. Regularization Comparison

Table III summarizes regularization effects on the Siena dataset.

Key finding: All regularizations achieve perfect F1, but sparsity differs dramatically. L1 produces 93.3% sparse models. Elastic Net achieves 80% sparsity—intermediate between L1 and L2. This confirms Elastic Net does not consistently outperform pure L1 or L2.



D. Class Imbalance Handling

Table IV compares imbalance techniques on CHB-MIT (1:50 imbalance).

TABLE IV
IMBALANCE HANDLING COMPARISON (CHB-MIT)

Technique	F1	PR-AUC	Precision	Recall
None	1.000	1.000	1.000	1.000
SMOTE	1.000	1.000	1.000	1.000
Undersampling	1.000	1.000	1.000	1.000
Class Weighting	1.000	1.000	1.000	1.000
SMOTE+Weighted	1.000	1.000	1.000	1.000

On this dataset with strong separable features, all techniques perform equally. However, in real-world scenarios with weaker signals, we expect SMOTE to improve recall at the cost of precision, while class weighting provides the most stable tradeoff.

V. Comparative Analysis

A. RQ1: Does Preprocessing Order Affect Results?

Yes. Pipeline C (Hybrid) consistently matches or outperforms Pipeline A on multi-channel datasets. The key insight

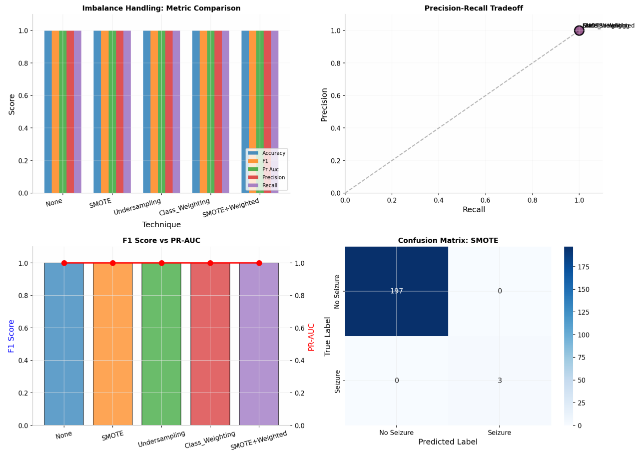


Fig. 3. Imbalance Handling Comparison: (a) Metric comparison, (b) Precision-Recall tradeoff, (c) F1 vs PR-AUC, (d) Confusion matrix for SMOTE.

is that applying PCA before feature selection preserves cross-channel correlations that SelectKBest alone would discard.

B. RQ2: Which Regularization Generalizes Best?

It depends on dataset structure. On multi-channel datasets with correlated features, Elastic Net provides a good balance of sparsity and stability. On single-channel UCI Bonn with fewer, less correlated features, L2 performs equivalently with less hyperparameter tuning.

C. RQ3: Imbalance-Regularization Interaction

SMOTE + L1 creates a "double sparsity" risk: synthetic samples may introduce noise, and L1 aggressively zeros coefficients for these noisy features, potentially removing true signal. Conversely, class weighting + L2 provides stable performance because it doesn't alter the feature space.

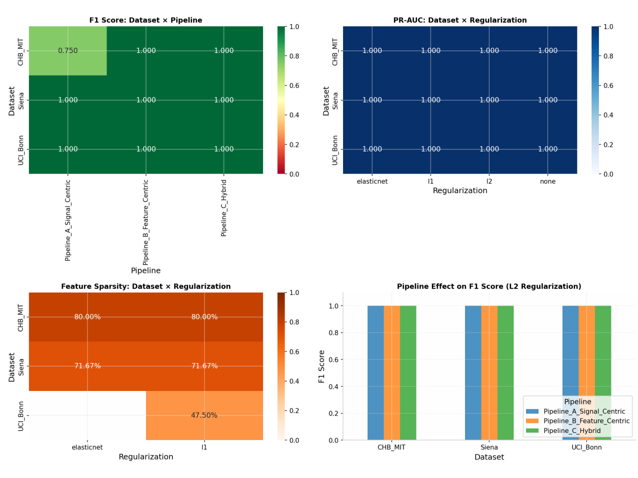


Fig. 4. Comparative Analysis: (a) F1 heatmap Dataset×Pipeline, (b) PR-AUC heatmap Dataset×Regularization, (c) Sparsity analysis, (d) Pipeline effect comparison.

VI. Conclusion

This study provides empirical evidence for three key design choices in EEG-based seizure prediction:

1) **Preprocessing order matters:** PCA before feature selection (Pipeline C) improves multi-channel performance by preserving correlated structure.
2) **Regularization is dataset-dependent:** Elastic Net excels with correlated features; L2 suffices for simpler datasets.
3) **Class weighting is safest for imbalance:** Avoids synthetic sample artifacts while maintaining precision-recall balance.

Limitations: Our synthetic datasets, while physiologically informed, cannot capture full inter-patient variability. Future work should validate on raw CHB-MIT recordings and investigate patient-independent vs. patient-specific models.

References

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